

MG2008-DATA ANALYSIS FOR BUSINESS-I (PROJECT OUTLINE)

Objective

- To apply statistical concepts in real scenarios
- To understand and interpret the meaning of data
- To learn about providing descriptive statistics
- To develop understanding of statistical techniques
- To learn about providing descriptive statistics
- To understand how to conduct research projects

General Description

You should have an idea of your research question i.e. what are you addressing in your project? You need to investigate a problem, impact, trend or relationship between variables and testing the claims. Good projects begin with clear & well defined objectives accordingly.

These steps that need to be followed:

- Think of the objectives of research in which you may be interested to investigate.
- Keep in mind that you collect a real data only and give some evidence of your working either in a form of visiting card of a person/email or any other authentic way/video/selfies for mall/University/company/bank survey etc
- Each group will collect a sample of size at least 50.
- Request the person for his/her time to fill the questionnaire and be gratified to the person to fill the questionnaire. Behave professionally when you visit the companies and follow formal dress code.
- Each student will go for at least 10 valid questionnaires.
- Compile data in excel sheet and run analysis using excel functions
- Submit your data file with your section and group name/topic for the folder
- Collect information relevant to the topic. It may be used in the description of your topic and defining your research questions. Here you will explain the variables under study. Information may be collected from books, journals, magazines (only credible sources as you will also have to reference them).
- Synthesize your basic understanding of the topic and formulate research question(s).
- Based on the research question(s), Perform data analysis, present findings and draw conclusions/decision making. Discuss appropriate cross comparisons.

Project Report:

Your group should hand in one project by the due date given. The project report should include the following:

- Introduction: the introduction section of your report should be as specific as possible, describing
- topic of your project
- main issues or problems you plan to address
- Research Questions/Objectives/Hypotheses
- source of information (from where you obtained the background information)
- data that you plan on using, including the variables measured

Data Analysis:

After gathering data using questionnaire, you have to perform data analysis (whole sample size allotted to you). Then forward the analysis/discussion according to your specified objectives. You'll have to run the following functions and submit a detailed write up of all the findings under separate headings. For each part of the analysis you will present the analysis in the form of tables/graphs and also explain the analysis for example maximums, minimums, trends, increase or decrease, differences etc.

- Application of relevant techniques (state your reasons why the specific measures are applied). The techniques should be relevant to your topic and make appropriate comparisons.

- Nature of the variables & measurement scales followed by each
- Descriptive Statistics

Frequencies, Percentages, Cumulative percentages/frequencies, measures of central tendency & Dispersion, comparisons, cross-tabulation and so on. All we have studied. (Apply only appropriate measures according to your data/questions). Provide descriptive statistics for all questions/demographics in an appropriate manner.

- *Graphs*: Bar, Pie, histogram, stem and leaf, line graphs or any other based on the relevance of the topic.
- *Shape of the Distribution*: comment if there is skewness or normalization. Discuss.
- *Outlier Detections*: using z-scores or Bar chart
- *Probability*
- *Perform correlation and regression analysis, Index numbers, Probability tables (if/where appropriate).*
- *Interpretation* of the overall results obtained.

Note: (state your reasons why the specific measures/tests are applied). The techniques should be relevant to your topic and make appropriate comparisons if necessary.

Conclusion: This will include general conclusions about the findings summing up everything in the end. Give Justifications which are sufficient to meet your objectives. You can also list down the limitations or problems you faced while conducting this research.

References: Under this section, provide all information about the online resources you took help from. (Both the information used to explain the topic and the source(s) from where you obtained data) Use APA referencing style.

Appendix: All the **questionnaires, data files** should be attached in **the appendix section** of the report

Note: You are expected to submit a good quality report where everything stated is well written and well presented. Also, you are supposed to **submit both soft and hard copies** of the report.

Project Submission:

Due Date: 4th May 2026.

Students are required to submit both hard and soft copy of their Project Reports. Project will not be considered for grading unless submitted in both formats.

Note: The project deliverable and presentation (If slots are available) will be conducted at the end of the semester. Late submissions will automatically mean a 2% penalty on each deliverable hence, do not complain afterwards.

Hard Copy Submission:

- Use colour separators of each section and tape bind the report.
- Report should have a title page which is attached.
- Report should be in Portrait format.
- Font: **Times New Roman**, Font size: **Headings=14, All text=12**
- Page Size: **A4**, Page Margins: **Left=1, Top=0.75, Bottom=0.75, Right=0.75**
- Report should have the following sequence of chapters/sections and subsections

ABSTRACT new page (Separator - 1)

ACKNOWLEDGEMENT new page (Separator - 1)

TABLE OF CONTENTS new page (page number starts from Introduction followed a Style: 1|Page)(Separator - 2)

LIST OF TABLES new page(Separator - 2)

LIST OF FIGURES same/new page(Separator - 2)

LIST OF ABBREVIATIONS same/new page(Separator - 2)

1. INTRODUCTION & LITERATURE new page (Separator - 3)

2. 1.1. OBJECTIVES OF THE STUDY new page(Separator -4)

3. DATA COLLECTION & METHODOLOGY new page(Separator - 5)

4. ANALYSIS & RESULTS new page(Separator - 6)

4.1. Graphical Summaries new page

4.2. Tabular Summaries new page

4.3. So on.

5. **CONCLUSION/DISCUSSION** new page(*Separator - 7*)

6. **LIMITATIONS OF THE STUDY** new page(*Separator - 7*)

7. **REFERENCES**(*Separator - 7*)

8. **EVIDENCE OF WORKING**(*Separator - 8*)

A) (PICS/CARDS)

B) SPECIMEN OF QUESTIONNAIRE

C) SAMPLE OF 5 FILLED QUESTIONNAIRES